

Sampling weight adjustments in partial least squares structural equation modeling: guidelines and illustrations

Jun-Hwa Cheah, José L. Roldán, Enrico Ciavolino, Hiram Ting & T. Ramayah

To cite this article: Jun-Hwa Cheah, José L. Roldán, Enrico Ciavolino, Hiram Ting & T. Ramayah (2021) Sampling weight adjustments in partial least squares structural equation modeling: guidelines and illustrations, *Total Quality Management & Business Excellence*, 32:13-14, 1594-1613, DOI: [10.1080/14783363.2020.1754125](https://doi.org/10.1080/14783363.2020.1754125)

To link to this article: <https://doi.org/10.1080/14783363.2020.1754125>



Published online: 27 Apr 2020.



Submit your article to this journal [↗](#)



Article views: 1550



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 29 View citing articles [↗](#)

Sampling weight adjustments in partial least squares structural equation modeling: guidelines and illustrations

Jun-Hwa Cheah^{a*}, José L. Roldán^b, Enrico Ciavolino ^c, Hiram Ting ^{de} and T. Ramayah^{fg}

^a*School of Business and Economics, Universiti Putra Malaysia, Serdang, Selangor, Malaysia;*

^b*Department of Business Management and Marketing, Universidad de Sevilla, Seville, Spain;*

^c*Department of History, Society and Human Studies, University of Salento, Lecce, Italy;* ^d*Faculty of Hospitality and Tourism Management, UCSI University, Sarawak, Malaysia;* ^e*Ming Chuan University, Taoyuan, Taiwan;* ^f*School of Management, Universiti Sains Malaysia, Penang, Malaysia;* ^g*Internet Innovation Research Center, Newhuadu Business School, Minjiang University, Fuzhou, People's Republic of China*

Applications of partial least squares structural equation modelling (PLS-SEM) often draw on survey data. While researchers go to great lengths to document reliability and validity statistics that support the generalisability of their findings, they often overlook or ignore a more fundamental issue related to data analysis—the representativeness of their sample. Addressing this concern, the present paper offers guidelines for using the weighted PLS-SEM (WPLS-SEM) algorithm to apply sampling weights in the model estimation. The results of the WPLS algorithm and the traditional PLS algorithm are then compared using a marketing research model. The findings show that researchers should routinely consider the procedure of the WPLS algorithm when using the PLS technique for assessment. The WPLS algorithm is a useful and practical approach for achieving better average population estimates in situations where researchers have a set of appropriate weights. This paper substantiates the use of the WPLS algorithm and provides business researchers and practitioners with the proper guidelines to assess, report, and interpret PLS-SEM results. It also illustrates that the use of the WPLS algorithm produces different inference test results in the structural model and different predictive relevance results. Thus, the study contributes to the advancement of PLS-SEM applications.

Keywords: marketing; sampling weights; post-stratification weights; weighted PLS (WPLS); PLS-SEM

Introduction

Partial least squares structural equation modelling (PLS-SEM) is a widely used method for analysing complex inter-relationships between observed and latent variables. While the method is applied in a variety of disciplines – including agricultural science, engineering, environmental science, medicine, and psychology (e.g. Avkiran, 2018; Willaby et al., 2015) – its main area of application is in business research, particularly information systems (Cheah et al., 2019; Hair et al., 2017a; Ringle et al., 2012; Roldán & Sánchez-Franco, 2012; Shiau & Chau, 2016; Urbach & Ahlemann, 2010) and marketing (Hair et al., 2012; Hair et al., 2017c; Henseler et al., 2009). For example, the unified theory of acceptance and use of technology model (UTAUT), arguably one the most widely used models for studying user adoption in information systems, is typically estimated using PLS-SEM (e.g. Venkatesh et al., 2003; Venkatesh et al., 2012). Similarly, marketing scholars and

*Corresponding author. Email: jackycheahjh@gmail.com

practitioners use PLS-SEM for the American Customer Satisfaction Index (Fornell et al., 1996), the European Performance Satisfaction Index (Westlund et al., 2008), and the European Customer Satisfaction Index (Cassel et al., 2000; Eklöf & Westlund, 2002; Hsu et al., 2006; O'Loughlin & Coenders, 2004) – three indices that have been extensively adopted in research and practice alike.

Practically all studies that use PLS-SEM frame their approach in a confirmatory sense – that is, a literature review is followed by the derivation of formal hypotheses and finally the model estimation (Henseler, 2018). At the same time, however, researchers emphasise the predictive nature of their results when deriving managerial implications (Hair et al., 2019b; Tenenhaus, 2008). While this *modus operandi* conforms to PLS-SEM's 'causal–predictive' nature (Jöreskog & Wold, 1982, p. 270), it also places specific requirements on the research design, particularly with regard to data collection. Most notably, depending on whether the research aims are to test a theory or make predictive statements, researchers have to ensure that a sample is representative of the population (Memon et al., 2017). More precisely, when testing a set of formal hypotheses (i.e. theory application), researchers evaluate a specific theory on effects beyond a single research setting (Lichters et al., 2015). Hence, 'effects observed in the research are employed to assess the status of theory. Nevertheless, it is the theoretical explanation that is expected to be generalizable and not the particular effects obtained' (Calder et al., 1981, p. 197). In this case, sample representativeness is of secondary concern when comparing models, as long as specific sample characteristics are not an integral part of the theory (Calder et al., 1981). This differs from effects application, which, according to Calder et al. (1981, p. 198),

is based on a desire for knowledge about the events and relationships in a particular real-world situation. The primary goal of this type of research (...) is to obtain findings that can be applied directly to the situation of interest.

In such a setting, researchers need to ensure that their sample is representative of the specific population of interest (Lichters et al., 2015). As deriving managerial implications has become a *sine qua non* for publishing articles in business research journals, including that of marketing, ensuring sample representativeness should not be overlooked or ignored in any PLS-SEM analysis.

As gathering a representative sample tends to be prohibitively expensive or even impossible (Cooke, 2017), most researchers apply non-probability sampling methods such as convenience sampling (Sarstedt et al., 2018a). These sampling methods, however, generally lack representativeness and generalisability (e.g. Levy & Lemeshow, 2013). But even if researchers use more sophisticated sampling methods, issues such as unequal selection probabilities, non-response, and non-coverage pose a threat to a sample's representativeness (Kalton & Flores-Cervantes, 2003). To address these limitations, researchers can assign sampling weights (also known as post-stratification weights) to each observation in order to ensure that the weighted observations represent the population of interest as closely as practically possible (e.g. Groves et al., 2002).¹ Statistical procedures then process these sampling weights in the computation of results. For example, if a population consists of an equal share of males and females but the data collected comprises 60% males and 40% females, then the use of the sampling weights would ensure that females are weighted equal to males in the result analysis.

A corresponding extension of the original PLS-SEM algorithm has recently been introduced by Becker and Ismail (2016). Their WPLS-SEM algorithm allows researchers to specify a weighting vector that defines the relevance of each observation for the computation of results. With the WPLS-SEM algorithm now implemented in standard PLS-SEM

software, such as SmartPLS (Ringle et al., 2015), researchers can readily use the procedure. However, despite its obvious importance, researchers have not utilised this methodological extension yet. As of January 2020, the paper by Becker and Ismail (2016) has been cited only 24 times in *Google Scholar*. Specifically, the paper has been cited in articles that have reviewed PLS-SEM use in different disciplines (e.g. Ali et al., 2018; Richter et al., 2015) or to support the method's use in a specific context (e.g. Galindo-Martín et al., 2019). Not a single study applied the WPLS-SEM algorithm. One potential reason is a lack of guidance and thus understanding among researchers on how to use the WPLS-SEM algorithm.

Addressing these concerns, this paper offers a systematic explanation of the steps that researchers should follow when applying the WPLS-SEM algorithm. We illustrate the use of the WPLS-SEM algorithm with a marketing model and contrast its results with those produced by the original PLS-SEM algorithm. In doing so, we highlight the differences in the findings and conclusions when using the WPLS-SEM algorithm in a specific context. Our overarching aim is to encourage routine ex-post adjustments of the sampling weights in PLS-SEM analyses when conducting business studies related to behavioural issues.

The WPLS-SEM algorithm: procedure and guidelines

PLS-SEM is a system of iterative regressions that processes standardised indicator data to calculate weighted composites as approximations of the conceptual latent variables under consideration (Hair et al., 2019a). In the first stage, the PLS-SEM algorithm iteratively determines the inner weights and latent variable scores by means of a four-step procedure. Consistent with the algorithm's original presentation (Lohmöller, 1989), inner weights refer to path coefficients, while outer weights and outer loadings refer to indicator weights and loadings in the measurement models. In the second and third stages, the process uses the final latent variable scores from Stage 1 as the inputs for a series of ordinary least squares regressions. These regressions produce the final estimates of the outer loadings, outer weights, and path coefficients as well as related elements such as indirect and total effects, R^2 values of the endogenous latent variables, and the indicator and latent variable correlations (Lohmöller, 1989). However, such a procedure creates the difficulty of incorporating the sample weight because the PLS-SEM relies highly on standardised data during its iterative algorithm to form the composite scores.² Consequently, a precomputation sampling weight might not work because the resulting data set is used for the calculation of weighted composites as proxies for the conceptual latent variables. Standardising a pre-weighted data set will obscure the underlying correlation pattern because multiplication with a non-constant value is not commutative.

Nevertheless, Becker and Ismail (2016) have proposed an appropriate correction of the PLS-SEM algorithm when researchers are interested to incorporate the sampling weights. Instead of treating each observation equally, the WPLS-SEM algorithm takes weighting vectors into account for each calculation of mean, variance, and covariance (correlation) of the composites. In other words, when the basic PLS-SEM algorithm performs a standardisation of a variable each time, it uses the weighted mean and the weighted variance of the variable rather than the normal mean and the normal variance. Thus, the WPLS-SEM algorithm produces weighted means and the weighted (co)variances of the indicator variables over the iterations (see Becker & Ismail, 2016, equations (3)–(5), p. 609). From these statistics, the weighted covariance, the weighted standard deviations, and the weighted standardised beta coefficient are then used in the computation of the correlation weights (Mode A) and the regression weights (Mode B), depending on the measurement model specifications

(Becker et al., 2013; Cheah et al., 2018; Sarstedt et al., 2016).³ Similarly, the estimation of structural model relationships produces the weighted standardised path coefficients (Becker & Ismail, 2016, equation (6), p. 609). Overall, the effect of these corrections is that all the calculations during the iterative PLS algorithm (e.g. estimation of outer weights, outer loadings, and inner weights) are appropriately weighted with the sampling weights, while retaining all information from the original data set in the model.

To implement the WPLS-SEM procedure, researchers need to follow a four-step procedure as outlined in Figure 1.

Step 1: choose the right auxiliary variable for sampling weights

To calculate sampling weights, researchers need appropriate and meaningful auxiliary data to which the sample data can be compared (e.g. Little, 1993). In this vein, researchers need sufficient knowledge about the use of suitable auxiliary variables and how to define the population precisely, which is a prerequisite for calculating the weights. Some of the examples of auxiliary data that could be used to calculate sampling weights (also known as respondent characteristics) are gender, age, education, race/ethnicity, region, or place of residence (Kulas et al., 2018). To illustrate that point, a university knows from the registrations how many of its students are male and how many are female. Researchers can draw a sample from the student population and compare their sample to the data from the university. Moreover, according to Cameron and Trivedi (2005), more complex sampling weights can be created based on complex survey designs. For instance, assuming that disproportionate numbers in terms of gender appear in some of the age groups (i.e. disproportionate numbers of gender and age simultaneously), researchers could adjust the sampling weights by using multiple population characteristics.

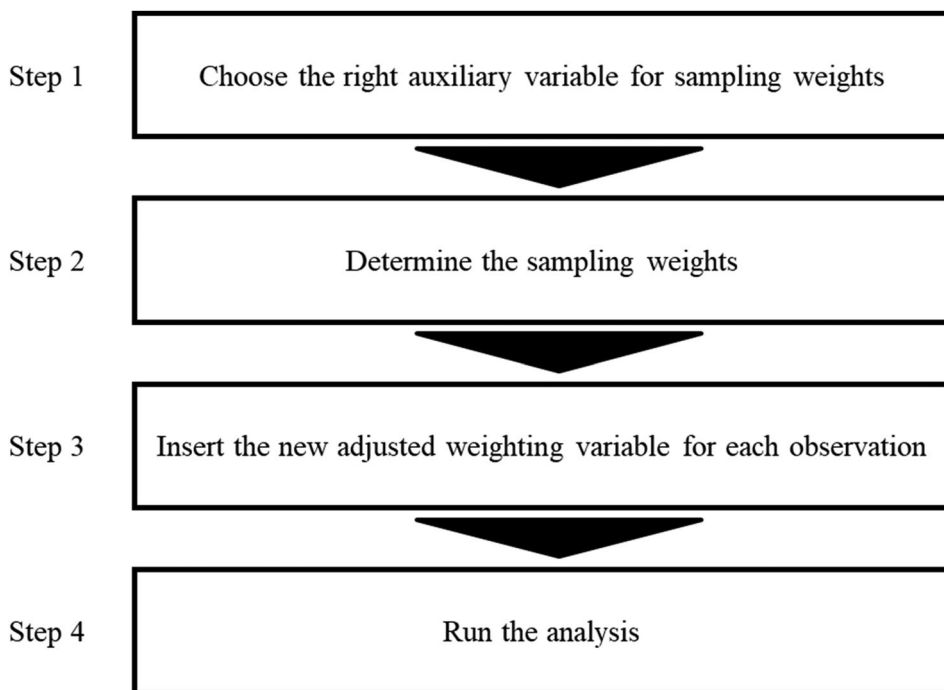


Figure 1. Steps of the WPLS-SEM procedure.

If the auxiliary variable is not correctly defined or the sampling procedures are inadequate, it could introduce a substantial design effect on the researcher data. Consequently, the issue of large variability in the computation of the weights will occur, and this can create an inefficient estimator without reducing bias (Bollen et al., 2016). In other words, poor sampling weights will increase standard errors in the analysis (i.e. PLS-SEM), thus making it less precise in inference estimation. Hence, the construction and use of sampling weights should follow a rigorous process and fulfil the basic requirements and assumptions (e.g. Becker & Ismail, 2016; Cameron & Trivedi, 2005; Gelman, 2007; Shin, 2012). Some of the key requirements to consider before using sampling weights are listed below:

- Researchers need to check (e.g. with crosstabs) whether their variable of interest is significantly related to the weighting variable (e.g. gender). If the variable of interest and potential weighting variable are not correlated, there is no need to weight because the weighting variable does not add any valuable information. For example, if the outcome variable (i.e. time spent studying) has nothing to do with gender, it will not help in fixing the unequal selection probabilities because it has no additional information (Toepoel, 2015).
- Any auxiliary variables for sampling weight usage must not be related to response or non-response. For instance, in an online survey, age can be a tricky weighting variable because response probabilities are often associated with age – especially the old have lower response probabilities compared to the young (Toepoel, 2015).
- The auxiliary variables in a sample need to be measured in the same way as those for the population. This requirement is usually easy to fulfil with variables like age or gender. However, it can be complicated when the constructs under investigation have multiple definitions, such as employment status, occupation classification, or small and medium-sized enterprises (SMEs). For instance, SMEs in the Malaysia manufacturing sector are defined as firms with a sales turnover not exceeding RM50 million or a number of full-time employees not exceeding 200 (National SME Development Council, 2013). However, if researchers plan to collect data on services and other sectors, the current definition of an SME in the manufacturing sector would not be suitable to split the sample into the appropriate cells. According to the National SME Development Council (2013), SMEs in services and other sectors can be defined as firms with sales turnover not exceeding RM20 million and with less than 75 full-time employees. From this definition, researchers may not group the sample based on the right definition of the population. Therefore, careful assessment of the equivalence of the measured variables and their definitions is necessary when using them for weighting purposes.
- The auxiliary variables that define the weights should not include the endogenous variable. For example, if income is an endogenous variable in the model, it should not be used to define the sampling weights that correct underrepresentation in terms of the income distribution. Using the dependent variable for weighting would bias the results.
- Even if the data originates from a composite model population, unbiased and efficient estimates cannot be expected from a non-probability sample. Thus, the approach requires selecting the sample elements in each cell at random (Shin, 2012).
- The sampling weights should not be used as ‘fishing expeditions’ or ‘*p* hacking’ to achieve appealing inference estimation (also known as data manipulation) (Becker & Ismail, 2016).

- If the values of the sampling weights are the same for all individuals within the group, the weights are not needed in a WPLS-SEM analysis (Groves et al., 2002).
- There should be an adequate number of sample elements in all cells to allow the efficient estimation of the respective parameters. Single sampling units' idiosyncrasies might therefore be multiplied, and the parameter estimates might be worse than ignoring the sampling disproportionate number. The observations within a cell also need to fulfil the minimum requirements of sample size. The minimum within-cell sample size, recommended by Westfall et al. (2011), should be at least 10 observations for stability of the mean and the estimated standard error of the mean. Furthermore, Kalton (1983) highlighted that the number of respondents in an all-weighting cell should be reasonably large – that is, the observation should be larger than or equal to 20. Hence, researchers need to have clear frequency or crosstab tables to ensure the number of cases in each cell in the sample is not too small (i.e. Sarstedt & Mooi, 2019, see Chapter 5). In contrast, the strategy of oversampling is often applied to increase the precision of the estimation for a group with only a few population members.

Step 2: determine the sampling weights.

The WPLS-SEM algorithm draws on a weighting variable that defines a sampling weight for every observation in the dataset. To determine each observation's sampling weight, the sample structure is compared with the population structure for each auxiliary variable. With this aim in mind, certain steps are followed: (1) A 'Proportion of the Population' is computed for the auxiliary variable or a combination of auxiliary variables. In this way, researchers should first use each of the member populations of each group generated divided by the total of the population. (2) A 'Proportion of the Sample' is calculated using each of the group samples divided by the total of the sample. (3) Finally, the 'Proportion of the Population' should be divided by the 'Proportion of the Sample' to generate the group 'Weight' results.

Step 3: insert the new adjusted weighting variable for each observation

Once the weights have been determined for each group of respondents, these values need to be inserted in the original dataset, which is then incorporated into the software selected to conduct the PLS-SEM analysis. To do so, researchers can use any statistical software programme to define the new adjusted weighting variable.⁴

Step 4: run the analysis.

A weighting variable should be selected in the chosen PLS-SEM software, provided that the statistical programme allows for such a functionality. Next, that routine will be executed, and each observation will be weighted according to the sampling weights.

Case study

Our empirical illustration of the WPLS algorithm draws on data from Martín Ruiz et al.'s (2008) service value index model.⁵ This model measures respondents' perceived service value using four dimensions: service quality, service equity, confidence benefits, and perceived sacrifice. We use these dimensions as first-order constructs to explain respondents'

service satisfaction and repurchase intention (Taylor & Baker, 1994; Zeithaml et al., 1996). All constructs adopt multi-item reflective measurement models. Figure 2 illustrates the path model under consideration. The dataset includes a sample of 500 consumers who regularly dine at one of the five major fast food chains in Malaysia (Burger King, KFC, McDonald's, Pizza Hut, or Subway). See Cheah et al. (2018) for further details on the measurement instrument generation and sample characteristics.

We use the SmartPLS 3.2.9 (Ringle et al., 2015; Sarstedt & Cheah, 2019) software to estimate the model parameters. In line with Hair et al.'s (2017b) recommendations, we use path weighting, with a maximum of 300 iterations, and a stop criterion of 10⁻⁷ in the PLS-SEM algorithm settings.

Implementing the WPLS-SEM algorithm

Step 1: choose the right auxiliary variable for sampling weights

It is common for market researchers to use sampling weight techniques in survey data analysis (e.g. Biemer & Christ, 2008; Cameron & Trivedi, 2005). Since consumer behaviour surveys are universally affected by non-coverage error, many surveys will employ sampling weights or population control adjustments of age and gender as a statistical procedure to adjust survey results for non-coverage error (see Biemer & Christ, 2008). In addition, gender and age group are the most commonly used population characteristics in weighting (see Becker & Ismail, 2016). Drawing from the past literature, we decided to use gender and age as the auxiliary variables to incorporate the sampling weights.

We analyse the within-cell sample to guarantee an appropriate number of sample elements in all cells. For example, if a cell of males or females aged >55 years only has

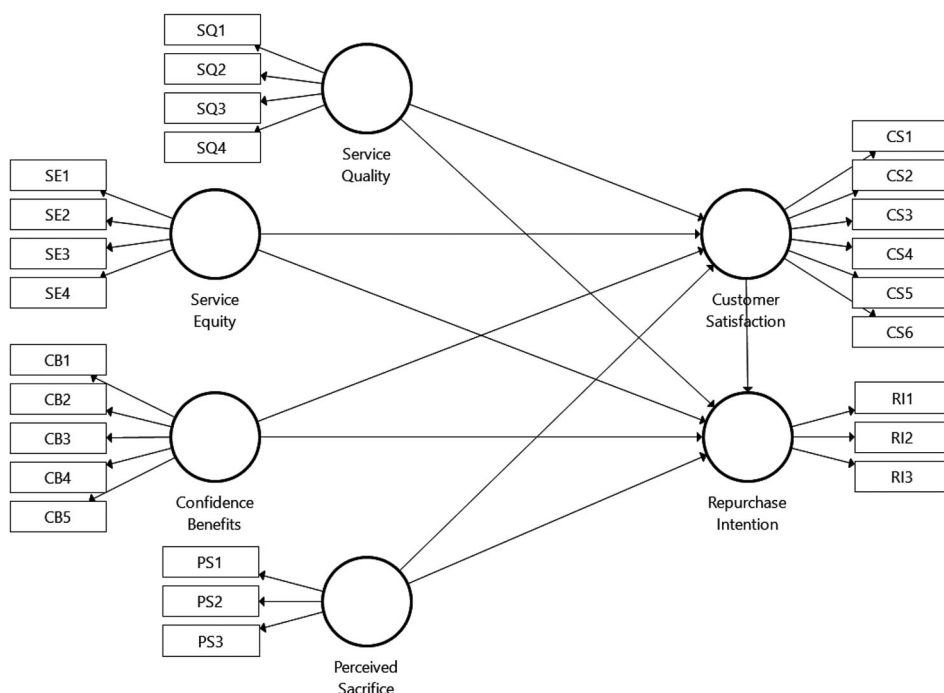


Figure 2. Empirical example of service value index model.

five observations, it could be assigned a very large weight. Based on Table 1, the result exhibits that the fourth cell in the sample (the cell of males and females within the age range of 45–54 years) is too small (two and seven observations). This could pose a problem in computing the weighting score. To avoid large weighting scores that could influence the parameter estimates, the present study excluded the cell of males and females within the age range of 45–54 years. Thus, our total sample size in the study was 491 ($500-9=491$).

Step 2: determine the sampling weights

With the aim of computing each observation's sampling weight, we compared our sample structure with the population structure of Kuala Lumpur. To do so, we obtained the distribution of age and gender of the Kuala Lumpur population from the Department of Statistics Malaysia (<https://www.dosm.gov.my/v1/>). First, to calculate the 'Proportion of the Population', we should first use each of the member populations of age group and gender divided by the total of the population. Second, to generate 'Proportion of the Sample', we used each of the group samples divided by the total of the sample. Finally, we obtained the group of the 'Weight' results on dividing the 'Proportion of the Population' by the 'Proportion of the Sample'. Table 2 contrasts the distributions of the age and gender groups for the sample and the population.

Next, researchers need to generate the weighting variable using the values defined in Table 2. For instance, Table 3 shows the syntax commands (see Sarstedt et al., 2018b) for generating the weighting variable using the values defined in Table 2. In the syntax, both age and gender are used to identify the groups of observations whose sampling weights need to be adjusted, whereas weight defines the weighting variable.

Step 3: insert the new adjusted weighting variable for each observation

After computing the weights for each group of respondents, we inserted these values into the original dataset. The extended dataset was then imported into the statistical software SmartPLS.

Table 1. Age*Gender cross tabulation.

			Gender		Total
			Male	Female	
Age	15–24	Count	51	165	216
		% within Age	23.60%	76.40%	100.00%
		% within Gender	35.90%	46.10%	43.20%
	25–34	Count	59	142	201
		% within Age	29.40%	70.60%	100.00%
		% within Gender	41.50%	39.70%	40.20%
	35–44	Count	30	44	74
		% within Age	40.50%	59.50%	100.00%
		% within Gender	21.10%	12.30%	14.80%
	45–54	Count	2	7	9
		% within Age	22.20%	77.80%	100.00%
		% within Gender	1.40%	2.00%	1.80%
Total	Count		142	358	500
	% within Age		28.40%	71.60%	100.00%
	% within Gender		100.00%	100.00%	100.00%

Table 2. Post-stratification weights of Kuala Lumpur census.

Age & Gender	Population <i>N</i>	Proportion of Population (PP)	Sample <i>n</i>	Proportion of Sample (PS)	Weight (PP/PS)
15–24: Male	2734859	0.173025	51	0.10387	1.666
15–24: Female	2880351	0.182229	165	0.336049	0.542
25–34: Male	2938372	0.185900	59	0.120163	1.547
25–34: Female	2886016	0.182588	142	0.289206	0.631
35–44: Male	2198785	0.139109	30	0.061100	2.277
35–44: Female	2167807	0.137149	44	0.089613	1.530
Total	15806190		491		

Table 3. SPSS syntax example.

```

IF (age = 1 AND gender = 1) weight = 1.666.
IF (age = 1 AND gender = 2) weight = 0.542.
IF (age = 2 AND gender = 1) weight = 1.547.
IF (age = 2 AND gender = 2) weight = 0.631.
IF (age = 3 AND gender = 1) weight = 2.277.
IF (age = 3 AND gender = 2) weight = 1.530.
EXE.

```

Step 4: run the analysis

To initiate the WPLS-SEM algorithm, SmartPLS offers the option to select a weighting variable in the PLS-SEM dialog box.⁶ To do this, we selected the Weighting tab, and specified the weighting variable (weight) under Weighting Vector. SmartPLS would then weigh each observation according to the sampling weights. The following sections document the results of the WPLS-SEM and the standard PLS-SEM algorithms.

Measurement model results

Following standard evaluation guidelines (Hair et al., 2017b, see Chapter 4), we first assessed the reliability and convergent validity of the reflectively specified measurement models. The results are presented in Table 4. The assessment of the indicator loadings shows that all the measures meet the threshold value of 0.7, except for the third item in perceived sacrifice (*PS3*) and the first item in repurchase intention (*R11*), which exhibit very low indicator reliabilities in both the WPLS-SEM and standard PLS-SEM analyses. Hence, we deleted these two items from the analysis. In addition, the first indicator of confidence benefits (*CB4*) in the standard PLS-SEM analysis and the second indicator of the same construct (*CB2*) in the WPLS-SEM analysis have loadings lower than 0.7. However, we retained these indicators because the constructs' composite reliabilities are above the recommended threshold of 0.70 (Hair et al., 2011). Likewise, the average variance extracted – which indicates the amount of variance that the construct captures in

Table 4. Measurement model results (WPLS-SEM and PLS-SEM).

Construct	Item	WPLS-SEM algorithm			PLS-SEM algorithm		
		Loading	CR	AVE	Loading	CR	AVE
Confidence Benefits	CB1	0.753	0.860	0.551	0.747	0.862	0.557
	CB2	0.727			0.730		
	CB3	0.737			0.761		
	CB4	0.674			0.684		
	CB5	0.813			0.802		
Customer Satisfaction	CS1	0.869	0.951	0.764	0.867	0.952	0.769
	CS2	0.885			0.882		
	CS3	0.867			0.883		
	CS4	0.868			0.868		
	CS5	0.901			0.903		
	CS6	0.853			0.859		
Perceived Sacrifice	PS1	0.838	0.826	0.703	0.816	0.823	0.700
	PS2	0.839			0.857		
	PS3	D			D		
Repurchase Intention	RI1	D	0.861	0.757	D	0.851	0.743
	RI2	0.804			0.788		
	RI3	0.931			0.929		
Service Equity	SE1	0.731	0.867	0.620	0.719	0.870	0.626
	SE2	0.810			0.811		
	SE3	0.837			0.839		
	SE4	0.767			0.791		
Service Quality	SQ1	0.810	0.900	0.693	0.811	0.905	0.704
	SQ2	0.802			0.821		
	SQ3	0.845			0.850		
	SQ4	0.872			0.872		

Note: D = Indicator deleted due to low loading; CR = Composite Reliability; AVE = Average Variance Extracted.

relation to the variance due to measurement error – is higher than the 0.5 threshold for all constructs in the WPLS-SEM and PLS-SEM analyses.

To assess the construct measures' discriminant validity, we applied Henseler et al.'s (2016) heterotrait–monotrait ratio of correlations (HTMT) statistic. The results in Table 5 show that most of the HTMT values were significantly lower than 0.9 (Franke & Sarstedt, 2019). The only exceptions were the HTMT values for the relationship between customer satisfaction and repurchase intention in the WPLS-SEM and PLS-SEM analyses, as both confidence intervals included the value of 0.9. However, as satisfaction and repurchase intention are conceptually highly similar, such a strong relationship was expected. Hence, we found support for the measures' discriminant validity.

Structural model results

Table 6 documents the structural model results of the WPLS-SEM and PLS-SEM analyses, which indicate that the WPLS-SEM analysis produced higher levels of explanatory power than the PLS-SEM analysis as evidenced by higher R^2 values. While the WPLS-SEM analysis produced an R^2 value of 0.559 in satisfaction, in PLS-SEM this value was only 0.544. However, the R^2 of repurchase intention in PLS-SEM (0.541) was greater compared with WPLS-SEM (0.537). Although the R^2 values differed between the two different algorithms, the f^2 effect sizes did not differ much in magnitude, except for the relationship

Table 5. Assessment of discriminant validity using HTMT.

Algorithm	Construct	1	2	3	4	5	6
WPLS-SEM	1. Confidence Benefits						
	2. Perceived Sacrifice	0.283 [0.119; 0.335]					
	3. Repurchase Intention	0.699 [0.674; 0.885]	0.247 [0.162; 0.444]				
	4. Satisfaction	0.771 [0.645; 0.858]	0.350 [0.130; 0.567]	0.853 [0.793; 0.978]			
	5. Service Equity	0.574 [0.437; 0.750]	0.093 [0.030; 0.144]	0.556 [0.398; 0.765]	0.555 [0.382; 0.712]		
	6. Service Quality	0.760 [0.647; 0.877]	0.271 [0.046; 0.310]	0.576 [0.549; 0.845]	0.688 [0.704; 0.871]	0.451 [0.344; 0.737]	
PLS-SEM	1. Confidence Benefits						
	2. Perceived Sacrifice	0.315 [0.178; 0.414]					
	3. Repurchase Intention	0.693 [0.605; 0.792]	0.263 [0.178; 0.364]				
	4. Satisfaction	0.752 [0.658; 0.822]	0.339 [0.197; 0.477]	0.861 [0.811; 0.922]			
	5. Service Equity	0.560 [0.471; 0.659]	0.068 [0.026; 0.074]	0.566 [0.458; 0.674]	0.543 [0.454; 0.636]		
	6. Service Quality	0.756 [0.665; 0.828]	0.256 [0.133; 0.343]	0.585 [0.505; 0.699]	0.696 [0.644; 0.769]	0.468 [0.386; 0.589]	

Note: The values in the brackets represent the lower and the upper bounds of the 95% confidence interval.

between confidence benefits and customer satisfaction (WPLS-SEM: 0.175 vs PLS-SEM: 0.140). As shown in Table 5, the f^2 effect sizes of the exogenous latent variables in the WPLS-SEM model and the PLS-SEM model have small to medium effects. Interestingly, the WPLS-SEM analysis produced higher Q^2 values, as obtained by blindfolding with an omission distance of 7, in both satisfaction (0.398 vs 0.389) and repurchase intention (0.369 vs 0.365).

Comparing the path coefficients across the two analyses, Table 6 shows that while the significances were similar, some of the point estimates varied considerably between the WPLS-SEM and PLS-SEM estimations. For example, confidence benefits, perceived sacrifice, and service equity have much stronger effects on satisfaction in the WPLS-SEM analysis (0.385, 0.128, and 0.201) compared to the PLS-SEM analysis (0.348, 0.117, and 0.195). In contrast, the effects of service quality were less pronounced in the WPLS-SEM analysis (0.271) compared to the PLS-SEM analysis (0.307). More importantly, both the WPLS-SEM and PLS-SEM clearly indicated the pronounced relevance of confidence benefits

Table 6. Assessment of structural model

Model	Relationship	PLS-SEM Algorithm				WPLS-SEM Algorithm			
		Path Coefficient	Percentile Bootstrap 95% CI		f^2	Path Coefficient	Percentile Bootstrap 95% CI		f^2
			LB	UB			LB	UB	
Path	Confidence Benefits → CS	0.348** (0.060)	0.218	0.453	0.140	0.385** (0.057)	0.269	0.492	0.175
	Perceived Sacrifice → CS	0.117** (0.036)	0.046	0.184	0.028	0.128** (0.036)	0.055	0.197	0.035
	Service Equity → CS	0.195** (0.037)	0.122	0.266	0.064	0.201** (0.041)	0.114	0.277	0.069
	Service Quality → CS	0.307** (0.057)	0.204	0.427	0.120	0.271** (0.054)	0.169	0.365	0.096
	Confidence Benefits → RI	0.100* (0.047)	0.007	0.192	0.010	0.090 (0.053)	-0.015	0.191	0.008
	Perceived Sacrifice → RI	-0.053 (0.037)	-0.125	0.020	0.006	-0.010 (0.040)	-0.090	0.067	0.000
	Service Equity → RI	0.092* (0.043)	0.008	0.175	0.013	0.083 (0.045)	-0.009	0.171	0.011
	Service Quality → RI	-0.016 (0.049)	-0.109	0.085	0.000	-0.004 (0.049)	-0.097	0.096	0.000
R^2	CS → RI	0.637** (0.045)	0.542	0.720	0.403	0.628** (0.052)	0.524	0.730	0.376
	CS	0.544				0.559			
Q^2	RI	0.541				0.537			
	CS	0.389				0.398			
	RI	0.365				0.369			

Note: ** $p < 0.01$, * $p < 0.05$ (one-tailed test); CS: Customer Satisfaction, RI: Repurchase Intention

and service quality as the most important drivers of customer satisfaction, closely followed by service equity and perceived sacrifice. However, there were significant differences in terms of the relationship of all the predictors (confidence benefits, perceived sacrifice, service equity, service quality, and customer satisfaction) with repurchase intention when comparing the WPLS-SEM and PLS-SEM analyses. The results show that the WPLS-SEM analysis produced a significant result of customer satisfaction for repurchase intention only. In contrast, PLS-SEM analysis produced three significant results of confidence benefits, service equity, and customer satisfaction for repurchase intention.

These divergences also translated into different conclusions when testing for potential mediating effects (Nitzl et al., 2016; Zhao et al., 2010). Specifically, in the WPLS-SEM analysis we found full mediation in all the relationships between the predictors (confidence benefits, perceived sacrifice, service equity, and service quality) and repurchase intention via satisfaction, whereas this was not the case in the PLS-SEM analysis (Table 7).

Overall, the WPLS-SEM analysis suggested a different prioritisation of the antecedent constructs and yielded different results in the mediation analysis, resulting in different conclusions about the mechanisms through which the antecedent constructs affected repurchase intentions.

In a final step, we compared the model estimates from the WPLS-SEM and PLS-SEM analyses with regard to their fit and predictive capabilities. For the first analysis, we computed Schwarz's (1978) Bayesian information criterion (BIC), which achieves a sound trade-off between model fit and predictive power in the estimation of the PLS-SEM technique (Sharma et al., 2019a; Sharma et al., 2019b).⁷ Based on Table 8, we found that the BIC criterion clearly favours the model estimated using WPLS-SEM when assuming

Table 7. Assessment of mediation result.

Algorithm	Relationship	Path Coefficient	BCa 95% CI		Type of Mediation
			LB	UB	
PLS-SEM	Confidence Benefits → Satisfaction → Repurchase Intention	0.222*** (0.041)	0.141	0.303	PM
	Perceived Sacrifice → Satisfaction → Repurchase Intention	0.074** (0.024)	0.028	0.123	FM
	Service Equity → Satisfaction → Repurchase Intention	0.124*** (0.026)	0.078	0.180	PM
	Service Quality → Satisfaction → Repurchase Intention	0.196*** (0.038)	0.127	0.276	FM
WPLS-SEM	Confidence Benefits → Satisfaction → Repurchase Intention	0.242*** (0.042)	0.162	0.328	FM
	Perceived Sacrifice → Satisfaction → Repurchase Intention	0.081*** (0.024)	0.035	0.126	FM
	Service Equity → Satisfaction → Repurchase Intention	0.126*** (0.027)	0.078	0.184	FM
	Service Quality → Satisfaction → Repurchase Intention	0.170*** (0.036)	0.105	0.247	FM

Note: *** $p < 0.01$, ** $p < 0.05$; FM: Full Mediation & PM: Partial Mediation.

Table 8. Model selection criteria and PLSpredict results (WPLS-SEM and PLS-SEM)

Model Selection Criteria	Customer Satisfaction		Repurchase Intention	
	PLS-SEM	WPLS-SEM	PLS-SEM	WPLS-SEM
BIC	-355.801	-372.444	-345.962	-341.892
	Customer Satisfaction		Repurchase Intention	
PLSpredict	PLS-SEM	WPLS-SEM	PLS-SEM	WPLS-SEM
RMSE	0.551	0.548	0.549	0.546
MAE	0.417	0.416	0.428	0.426
Q ² _predict	0.484	0.485	0.205	0.211

Note: Bold means represents better result on its criterion.

satisfaction as the target variable. In contrast, the BIC criterion result did not favour the model estimated with WPLS-SEM when assuming repurchase intention as the target variable, but the difference was not large at 4.092 (WPLS-SEM: -341.892 vs PLS-SEM: -345.962). In addition, to compare the different analyses with regard to their predictive power alone, we ran PLSpredict on the two models (Shmueli et al., 2016), using 10 folds and 10 repetitions (Shmueli et al., 2019).⁸ We found that for both target constructs, the WPLS-SEM analysis yielded slightly lower prediction errors in terms of mean absolute error (MAE) and root mean square error (RMSE) and high Q^2 _predict value (Table 8). Overall, our assessment of BIC and PLSpredict suggested that the model estimated using WPLS-SEM better fits the data and achieves a greater level of predictive power.

Discussion and recommendations

The accurate prediction of individuals' attitudes, perceptions, and behaviours depends, to a large degree, on the way researchers select participants from a defined population (Sarstedt et al., 2018a). Users of PLS-SEM must be critically aware of sample selection considerations, especially when they expect their findings to be applicable to the real world. However, drawing a sample that is representative of a population of interest is often infeasible. Moreover, issues such as unequal selection probabilities, non-response, and non-coverage complicate the achievement of representativeness in sampling (Kalton & Flores-Cervantes, 2003). To address this concern, researchers using PLS-SEM can engage in an ex-post adjustment of the sampling weights and apply Becker and Ismail's (2016) WPLS-SEM algorithm, which processes the sampling weights in the model estimation.

This paper aims to increase researchers' awareness and application of the WPLS-SEM algorithm by presenting some practical guidelines with a case study. Our comparison of model estimates from WPLS-SEM and standard PLS-SEM analyses does not disclose any differences in the results of the measurement model evaluation, which is consistent with the findings of Becker and Ismail (2016). However, our study shows that substantial differences could occur when the sampling weights are not taken into account in the structural model estimation. Such ignorance or negligence can affect the significance of the results and the magnitude of the path coefficient, as well as impair quality criteria, such as the explained variance (R^2) as well as the model's predictive relevance using the conventional blindfolding procedure (Q^2), the model selection criteria (BIC), and the PLSpredict (RMSE, MAE, Q^2 _predict). Furthermore, we found that the interpretation of the mediation effect could be distorted if the sampling inequalities are not appropriately addressed using the WPLS algorithm. Hence, we suggest that if researchers are interested in making

inferences to the population, they should correct the sampling deviations of their data set by using the WPLS algorithm to draw on less biased results or conclusions. Overall, our comparison thereby demonstrates the weighting of the sampling units in WPLS-SEM analysis and how they could yield findings different from a standard PLS-SEM analysis. This deviation could have serious repercussions on the theoretical and managerial implications drawn from the analysis and findings.

However, the impact of sampling weights is context-specific and depends on many factors, most notably on the sampling approach and population heterogeneity. For example, convenience sampling of a heterogeneous population will likely result in significant non-coverage, thereby increasing the impact of sampling weights. Hence, researchers should be careful when choosing the auxiliary variables which define the sampling weights. In particular, auxiliary variables should not include any dependent variable.

Finally, researchers need to ensure that the divergences between sampling and population distributions are not too pronounced. For example, in our empirical example, the divergence between sample and population distribution in the male age groups 35–44 produced high sampling weights (up to 2.277). Such high sampling weights can have a substantial impact on the result, particularly when the corresponding group of respondents has only a few observations. Hence, to eschew such bias, the observations within a group must fulfil the minimum requirements of the power analyses (e.g. Hair et al., 2017b; Marcoulides & Chin, 2013) or the minimum sample size generated from the inverse square root method as well as the gamma-exponential method (Kock & Hadaya, 2018).

Limitation and future studies

Overall, we hope that this paper will assist researchers to apply the WPLS algorithm in not only marketing studies but also other business research, such as that related to management and information systems. Furthermore, future research needs to extend the range of WPLS in different inference estimations. For example, the heterogeneity issue has become an increasing concern for marketing researchers. This refers to observations of a sample that can be assigned to segments a priori, on the basis of observable characteristics such as geographic or demographic traits (Hair et al., 2018, see Chapter 4). Hence, future research should explore the efficacy of the WPLS algorithm when estimating moderating effects in PLS-SEM, such as assessing using the interaction approach (Becker et al., 2018), multi-group analysis (Sarstedt et al., 2011), and measurement invariance of composite models (Henseler et al., 2016; Schlögel & Sarstedt, 2016).

Even though the WPLS algorithm has been incorporated in SmartPLS software at both the setting of the PLS algorithm and bootstrapping (Sarstedt & Cheah, 2019), future studies should extend this procedure as part of the weighted parameter estimation in PLSpredict (Shmueli et al., 2019). The ultimate aim is to continuously extend the SmartPLS toolbox of out-of-sample prediction analyses in PLS-SEM. Further research may also extend the existing knowledge on testing a research framework that includes both factor and composite models (Dijkstra & Henseler, 2015; Rigdon et al., 2017; Sarstedt et al., 2016) with the use of the WPLS algorithm. It can be useful to improve the statistical estimation and interpretation of path coefficients, outer model loadings, and outer model weights for both factor (using consistent PLS) and composite models (using Mode A or Mode B).

Finally, researchers should engage in different ex-post adjustments of sampling unit weights to compensate for critical issues in survey research, such as unequal selection probabilities, non-coverage, and sampling fluctuations (Bethlehem, 2010; Sarstedt et al., 2018a).

Numerous adjustment procedures, such as raking (e.g. Oh & Scheuren, 1983), linear weighting (e.g. Deville et al., 1993), and propensity score adjustment (e.g. Lee, 2006), all of which consider auxiliary information when adjusting sampling unit weights, have been proposed. See Kalton and Flores-Cervantes (2003) for an illustration of these procedures. As such, all these available adjustment procedures depend on the quality of the auxiliary information and the assumptions made by the researcher. The ultimate aim is to continuously extend the WPLS procedure in PLS-SEM by choosing the appropriate adjustment procedures to deal with survey research, such as unequal selection probabilities, non-coverage, and sampling fluctuations.

Acknowledgements

The authors thank Marko Sarstedt for his guiding comments and ideas regarding an earlier version of this manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes

1. The remainder of this article will only use the term ‘sampling weights’.
2. The composite scores have a trait of zero mean and standard deviation of one.
3. When using Mode A (i.e. correlation weights), the bivariate correlation between each indicator and the construct determines the outer weights. In contrast, Mode B (i.e. regression weights) computes indicator weights by regressing each construct on its associated indicators (Hair et al., 2017b).
4. Researchers can draw on the IF function in Microsoft Excel or use a statistical software program such as SPSS (Sarstedt & Mooi, 2019).
5. Cheah et al. (2018) use the same model and data in their comparative study on the use of single versus multiple items in PLS-SEM-based redundancy analyses.
6. The weighting can also be initiated in the context of the consistent PLS-SEM algorithm (Dijkstra & Henseler, 2015).
7. Note that Geweke and Meese’s (1981) criterion, which Sharma et al. (2019a, 2019b) also identified as suitable, is not applicable because our model is saturated.
8. The PLSpredict result for the PLS-SEM algorithm was estimated using SmartPLS (Ringle et al., 2015) while the PLSpredict result for the WPLS-SEM algorithm was estimated using the package <https://github.com/ISS-Analytics/pls-predict> for the statistical software R (Shmueli et al., 2016).

ORCID

Enrico Ciavolino  <http://orcid.org/0000-0003-3955-4310>

Hiram Ting  <http://orcid.org/0000-0002-1389-976X>

References

- Ali, F., Rasoolimanesh, S. M., Sarstedt, M., Ringle, C. M., & Ryu, K. (2018). An assessment of the use of partial least squares structural equation modeling (PLS-SEM) in hospitality research. *International Journal of Contemporary Hospitality Management*, 30(1), 514–538. <https://doi.org/10.1108/IJCHM-10-2016-0568>
- Avkiran, N. K. (2018). An in-depth discussion and illustration of partial least squares structural equation modeling in health care. *Health Care Management Science*, 21(3), 401–408. <https://doi.org/10.1007/s10729-017-9393-7>

- Becker, J.-M., & Ismail, I. R. (2016). Accounting for sampling weights in PLS path modeling: Simulations and empirical examples. *European Management Journal*, 34(6), 606–617. <https://doi.org/10.1016/j.emj.2016.06.009>
- Becker, J.-M., Rai, A., Ringle, C. M., & Völckner, F. (2013). Discovering unobserved heterogeneity in structural equation models to avert validity threats. *MIS Quarterly*, 37(3), 665–694. <https://doi.org/10.25300/MISQ/2013/37.3.01>
- Becker, J.-M., Ringle, C. M., & Sarstedt, M. (2018). Estimating moderating effects in PLS-SEM and PLSc-SEM: Interaction term generation*data treatment. *Journal of Applied Structural Equation Modeling*, 2(2), 1–21.
- Bethlehem, J. (2010). Selection bias in web surveys. *International Statistical Review*, 78(2), 161–188. <https://doi.org/10.1111/j.1751-5823.2010.00112.x>
- Biemer, P. P., & Christ, S. L. (2008). Weighting survey data. In E. D. de Leeuw, J. J. Hox, & D. A. Dillman (Eds.), *International handbook of survey methodology* (pp. 317–341). London: Routledge.
- Bollen, K. A., Biemer, P. P., Karr, A. F., Tueller, S., & Berzofsky, M. E. (2016). Are survey weights needed? A review of diagnostic tests in regression analysis. *Annual Review of Statistics and Its Application*, 3(1), 375–392. <https://doi.org/10.1146/annurev-statistics-011516-012958>
- Calder, B. J., Phillips, L. W., & Tybout, A. M. (1981). Designing research for application. *Journal of Consumer Research*, 8(2), 197–207. <https://doi.org/10.1086/208856>
- Cameron, A. C., & Trivedi, P. K. (2005). *Microeconometrics: Methods and applications*. Cambridge University Press.
- Cassel, C. M., Hackl, P., & Westlund, A. H. (2000). On measurement of intangible assets: A study of robustness of partial least squares. *Total Quality Management*, 11(7), 897–907. <https://doi.org/10.1080/09544120050135443>
- Cheah, J.-H., Sarstedt, M., Ringle, C. M., Ramayah, T., & Ting, H. (2018). Convergent validity assessment of formatively measured constructs in PLS-SEM: On using single-item versus multi-item measures in redundancy analyses. *International Journal of Contemporary Hospitality Management*, 30(11), 3192–3210. <https://doi.org/10.1108/IJCHM-10-2017-0649>
- Cheah, J.-H., Ting, H., Cham, T. H., & Memon, M. A. (2019). The effect of selfie promotion and celebrity endorsed advertisement on decision-making processes: A model comparison. *Internet Research*, 29(3), 552–577. <https://doi.org/10.1108/IntR-12-2017-0530>
- Cooke, M. (2017). Is technological change threatening the very existence of ‘traditional’ survey research and, if so, what should we do about it? *International Journal of Market Research*, 59(2), 153–155. <https://doi.org/10.2501/IJMR-2017-012>
- Deville, J.-C., Särndal, C.-E., & Sautory, O. (1993). Generalized raking procedures in survey sampling. *Journal of the American Statistical Association*, 88(423), 1013–1020. <https://doi.org/10.1080/01621459.1993.10476369>
- Dijkstra, T. K., & Henseler, J. (2015). Consistent partial least squares path modeling. *MIS Quarterly*, 39(2), 297–316. <https://doi.org/10.25300/MISQ/2015/39.2.02>
- Eklöf, J. A., & Westlund, A. H. (2002). The pan-European customer satisfaction index programme—current work and the way ahead. *Total Quality Management*, 13(8), 1099–1106. <https://doi.org/10.1080/095441202000000005>
- Fornell, C., Johnson, M. D., Anderson, E. W., Cha, J., & Bryant, B. E. (1996). The American customer satisfaction index: Nature, purpose, and findings. *Journal of Marketing*, 60(4), 7–18. <https://doi.org/10.2307/1251898>
- Franke, G. R., & Sarstedt, M. (2019). Heuristics versus statistics in discriminant validity testing: A comparison of four procedures. *Internet Research*, 29(3), 430–447. <https://doi.org/10.1108/IntR-12-2017-0515>
- Galindo-Martín, M.-Á., Castaño-Martínez, M.-S., & Méndez-Picazo, M.-T. (2019). Digital transformation, digital dividends and entrepreneurship: A quantitative analysis. *Journal of Business Research*, 101, 522–527. <https://doi.org/10.1016/j.jbusres.2018.12.014>
- Gelman, A. (2007). Struggles with survey weighting and regression modeling. *Statistical Science*, 22(2), 153–164. <https://doi.org/10.1214/0883423060000000691>
- Geweke, J., & Meese, R. (1981). Estimating regression models of finite but unknown order. *International Economic Review*, 22(1), 55–70. <https://doi.org/10.2307/2526135>
- Groves, R. M., Dillman, D. A., Eltinge, J. L., & Little, R. J. A. (2002). *Survey nonresponse*. New York: John Wiley & Sons.

- Hair, J., Hollingsworth, C. L., Randolph, A. B., & Chong, A. Y. L. (2017a). An updated and expanded assessment of PLS-SEM in information systems research. *Industrial Management & Data Systems*, 117(3), 442–458. <https://doi.org/10.1108/IMDS-04-2016-0130>
- Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017b). *A primer on partial least squares structural equation modeling (PLS-SEM)* (2nd ed.). Thousand Oaks: Sage.
- Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., & Thiele, K. O. (2017c). Mirror, mirror on the wall: A comparative evaluation of composite-based structural equation modeling methods. *Journal of the Academy of Marketing Science*, 45(5), 616–632. <https://doi.org/10.1007/s11747-017-0517-x>
- Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, 19(2), 139–151. <https://doi.org/10.2753/MTP1069-6679190202>
- Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019a). When to use and how to report the results of PLS-SEM. *European Business Review*, 31(1), 2–24. <https://doi.org/10.1108/EBR-11-2018-0203>
- Hair, J. F., Sarstedt, M., Ringle, C. M., & Gudergan, S. P. (2018). *Advanced issues in partial least squares structural equation modeling (PLS-SEM)*. Thousand Oaks: Sage.
- Hair, J. F., Sarstedt, M., & Ringle, C. M. (2019b). Rethinking some of the rethinking of partial least squares. *European Journal of Marketing*, 53(4), 566–584. <https://doi.org/10.1108/EJM-10-2018-0665>
- Hair, J. F., Sarstedt, M., Ringle, C. M., & Mena, J. A. (2012). An assessment of the use of partial least squares structural equation modeling in marketing research. *Journal of the Academy of Marketing Science*, 40(3), 414–433. <https://doi.org/10.1007/s11747-011-0261-6>
- Henseler, J. (2018). Partial least squares path modeling: Quo vadis? *Quality & Quantity*, 52(1), 1–8. <https://doi.org/10.1007/s11135-018-0689-6>
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. *International Marketing Review*, 33(3), 405–431. <https://doi.org/10.1108/IMR-09-2014-0304>
- Henseler, J., Ringle, C. M., & Sinkovics, R. (2009). The use of partial least squares path modeling in international marketing. In R. Sinkovics, & P. Ghauri (Eds.), *New challenges to international marketing (Advances in international marketing, vol. 20)* (pp. 277–319). Emerald Group Publishing Limited.
- Hsu, S. H., Chen, W. H., & Hsieh, M. J. (2006). Robustness testing of PLS, LISREL, EQS and ANN-based SEM for measuring customer satisfaction. *Total Quality Management & Business Excellence*, 17(3), 355–372. <https://doi.org/10.1080/14783360500451465>
- Jöreskog, K. G., & Wold, H. O. A. (1982). *The ML and PLS techniques for modeling with latent variables: Historical and comparative aspects*. North-Holland Publishing Company.
- Kalton, G. (1983). Models in the practice of survey sampling. *International Statistical Review/Revue Internationale de Statistique*, 51, 175–188.
- Kalton, G., & Flores-Cervantes, I. (2003). Weighting methods. *Journal of Official Statistics*, 19(2), 81. <https://doi.org/10.4236/csta.2017.62002> 770
- Kock, N., & Hadaya, P. (2018). Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods. *Information Systems Journal*, 28(1), 227–261. <https://doi.org/10.1111/isj.12131>
- Kulas, J. T., Robinson, D. H., Smith, J. A., & Kellar, D. Z. (2018). Post stratification weighting in organizational surveys: A cross disciplinary tutorial. *Human Resource Management*, 57(2), 419–436. <https://doi.org/10.1002/hrm.21796>
- Lee, S. (2006). Propensity score adjustment as a weighting scheme for volunteer panel web surveys. *Journal of Official Statistics*, 22(2), 329.
- Levy, P. S., & Lemeshow, S. (2013). *Sampling of populations: Methods and applications*. John Wiley & Sons.
- Lichters, M., Sarstedt, M., & Vogt, B. (2015). On the practical relevance of the attraction effect: A cautionary note and guidelines for context effect experiments. *AMS Review*, 5(1–2), 1–19. <https://doi.org/10.1007/s13162-015-0066-8>
- Little, R. J. (1993). Post-stratification: A modeler's perspective. *Journal of the American Statistical Association*, 88(423), 1001–1012. <https://doi.org/10.1080/01621459.1993.10476368>
- Lohmöller, J.-B. (1989). *Latent variable path modeling with partial least squares*. Heidelberg: Physica.

- Marcoulides, G. A., & Chin, W. W. (2013). You write, but others read: Common methodological misunderstandings in PLS and related methods. In H. Abdi, W. W. Chin, V. Esposito Vinzi, G. Russolillo, & L. Trinchera (Eds.), *New perspectives in partial least squares and related methods* (Vol. 56, pp. 31–64). Springer proceedings in mathematics & statistics. New York: Springer.
- Memon, M., Ting, H., Ramayah, T., Chuah, F., & Cheah, J. H. (2017). A review of the methodological misconceptions and guidelines related to the application of structural equation modeling: A Malaysian scenario. *Journal of Applied Structural Equation Modeling*, 1(1), 1–13.
- National SME Development Council. (2013, October). *Guidelines for new SME definition*. SME Corp. http://www.smecorp.gov.my/images/pdf/Guideline_New_SME_Definition_updated.pdf
- Nitzl, C., Roldán, J. L., & Cepeda, G. (2016). Mediation analysis in partial least squares path modeling: Helping researchers discuss more sophisticated models. *Industrial Management & Data Systems*, 116(9), 1849–1864. <https://doi.org/10.1108/IMDS-07-2015-0302>
- Oh, H., & Scheuren, F. (1983). Weighting adjustment for unit non-response. In W. G. Madow, I. Olkin, & D. B. Rubin (Eds.), *Incomplete data in sample surveys* (pp. 143–184). Academic Press.
- O’Loughlin, C., & Coenders, G. (2004). Estimation of the European customer satisfaction index: Maximum likelihood versus partial least squares. *Application to Postal Services. Total Quality Management & Business Excellence*, 15(9–10), 1231–1255. <https://doi.org/10.1080/1478336042000255604>
- Richter, N. F., Cepeda, G., Roldán, J. L., & Ringle, C. M. (2015). European management research using partial least squares structural equation modeling (PLS-SEM). *European Management Journal*, 33(1), 1–3. <https://doi.org/10.1016/j.emj.2014.12.001>
- Rigdon, E. E., Sarstedt, M., & Ringle, C. M. (2017). On comparing results from CB-SEM and PLS-SEM. Five perspectives and five recommendations. *Marketing ZFP*, 39(3), 4–16. <https://doi.org/10.15358/0344-1369-2017-3-4>
- Ringle, C. M., Sarstedt, M., & Straub, D. W. (2012). Editor’s comments: A critical look at the use of PLS-SEM in ‘MIS Quarterly’. *MIS Quarterly*, 36(1), iii–xiv. <https://doi.org/10.2307/41410402>
- Ringle, C. M., Wende, S., & Becker, J.-M. (2015). *SmartPLS 3*. Bönningstedt, SmartPLS.
- Roldán, J. L., & Sánchez-Franco, M. J. (2012). Variance-based structural equation modeling: Guidelines for using partial least squares in information systems research. In M. Mora, O. Gelman, A. Steenkamp, & M. Raisinghami (Eds.), *Research Methodologies, Innovations and Philosophies in software systems engineering and information systems* (pp. 193–221). IGI Global. <https://doi.org/10.4018/978-1-4666-0179-6.ch010>
- Ruiz, D. M., Gremler, D. D., Washburn, J. H., & Carrión, G. C. (2008). Service value revisited: Specifying a higher-order, formative measure. *Journal of Business Research*, 61(12), 1278–1291. <https://doi.org/10.1016/j.jbusres.2008.01.015>
- Sarstedt, M., Bengart, P., Shaltoni, A. M., & Lehmann, S. (2018a). The use of sampling methods in advertising research: A gap between theory and practice. *International Journal of Advertising*, 37(4), 650–663. <https://doi.org/10.1080/02650487.2017.1348329>
- Sarstedt, M., & Cheah, J. H. (2019). Partial least squares structural equation modeling using SmartPLS: A software review. *Journal of Marketing Analytics*, 7(3), 196–202. <https://doi.org/10.1057/s41270-019-00058-3>
- Sarstedt, M., Hair, J. F., Ringle, C. M., Thiele, K. O., & Gudergan, S. P. (2016). Estimation issues with PLS and CBSEM: Where the bias lies!. *Journal of Business Research*, 69(10), 3998–4010. <https://doi.org/10.1016/j.jbusres.2016.06.007>
- Sarstedt, M., Henseler, J., & Ringle, C. M. (2011). Multigroup analysis in partial least squares (PLS) path modeling: Alternative methods and empirical results. In M. Sarstedt, M. Schwaiger, & C. Taylor (Eds.), *Measurement and research methods in international marketing* (Vol. 22, pp. 195–218). Advances in International Marketing. Bingley: Emerald Group Publishing Limited. [https://doi.org/10.1108/S1474-7979\(2011\)0000022012](https://doi.org/10.1108/S1474-7979(2011)0000022012)
- Sarstedt, M., & Mooi, E. A. (2019). *A concise guide to market research: The process, data, and methods using IBM SPSS statistics*. Springer.
- Sarstedt, M., Schütz, T., & Raithel, S. (2018b). *IBM SPSS syntax: Eine anwendungsorientierte Einführung*. Vahlen.
- Schlägel, C., & Sarstedt, M. (2016). Assessing the measurement invariance of the four-dimensional cultural intelligence scale across countries: A composite model approach. *European Management Journal*, 34(6), 633–649. <https://doi.org/10.1016/j.emj.2016.06.002>

- Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics*, 34(6), 461–464. <https://doi.org/10.1214/aos/1176344136>
- Sharma, P. N., Sarstedt, M., Shmueli, G., Kim, K. H., & Thiele, K. O. (2019a). PLS-based model selection: The role of alternative explanations in information systems research. *Journal of the Association for Information Systems*, 20(4), 346–397. <https://doi.org/10.17705/1jais.00538>
- Sharma, P. N., Shmueli, G., Sarstedt, M., Danks, N., & Ray, S. (2019b). Prediction-oriented model selection in partial least squares path modeling. *Decision Sciences*, <https://doi.org/10.1111/deci.12329>
- Shiau, W.-L., & Chau, P. Y. (2016). Understanding behavioral intention to use a cloud computing classroom: A multiple model comparison approach. *Information & Management*, 53(3), 355–365. <https://doi.org/10.1016/j.im.2015.10.004>
- Shin, H. C. (2012). *A cautionary note on post-stratification adjustment*. Section on survey research methods - Joint Statistical Meeting (JSM), San Diego, CA.
- Shmueli, G., Ray, S., Velasquez Estrada, J. M., & Chatla, S. B. (2016). The elephant in the room: Evaluating the predictive performance of PLS models. *Journal of Business Research*, 69(10), 4552–4564. <https://doi.org/10.1016/j.jbusres.2016.03.049>
- Shmueli, G., Sarstedt, M., Hair, J. F., Cheah, J. H., Ting, H., Vaithilingam, S., & Ringle, C. M. (2019). Predictive model assessment in PLS-SEM: Guidelines for using PLSpredict. *European Journal of Marketing*, 53(11), 2322–2347. <https://doi.org/10.1108/EJM-02-2019-0189>
- Taylor, S. A., & Baker, T. L. (1994). An assessment of the relationship between service quality and customer satisfaction. *Journal of Retailing*, 70(2), 163–178. [https://doi.org/10.1016/0022-4359\(94\)90013-2](https://doi.org/10.1016/0022-4359(94)90013-2)
- Tenenhaus, M. (2008). Component-based structural equation modeling. *Total Quality Management*, 19(7–8), 871–886. <https://doi.org/10.1080/14783360802159543>
- Toepoel, V. (2015). *Doing surveys online*. SAGE Publications Ltd.
- Urbach, N., & Ahlemann, F. (2010). Structural equation modeling in information systems research using partial least squares. *Journal of Information Technology Theory and Application*, 11(2), 5–40.
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478. <https://doi.org/10.2307/30036540>
- Venkatesh, V., Thong, J. Y., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly*, 36(1), 157–178. <https://doi.org/10.2307/41410412>
- Westfall, J. A., Patterson, P. L., & Coulston, J. W. (2011). Post-stratified estimation: Within-strata and total sample size recommendations. *Canadian Journal of Forest Research*, 41(5), 1130–1139. <https://doi.org/10.1139/x11-031>
- Westlund, A. H., Källström, M., & Parmler, J. (2008). SEM-based customer satisfaction measurement: On multicollinearity and robust PLS estimation. *Total Quality Management*, 19(7–8), 855–869. <https://doi.org/10.1080/14783360802159527>
- Willaby, H. W., Costa, D. S. J., Burns, B. D., MacCann, C., & Roberts, R. D. (2015). Testing complex models with small sample sizes: A historical overview and empirical demonstration of what partial least squares (PLS) can offer differential psychology. *Personality and Individual Differences*, 84, 73–78. <https://doi.org/10.1016/j.paid.2014.09.008>
- Zeithaml, V. A., Berry, L. L., & Parasuraman, A. (1996). The behavioral consequences of service quality. *The Journal of Marketing*, 60(2), 31–46. <https://doi.org/10.2307/1251929>
- Zhao, X., Lynch Jr. J. G., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, 37(2), 197–206. <https://doi.org/10.1086/651257>